

MAT 2305- Real Analysis II
Assignment 01

Answers all the questions and submit the **solutions** for **1, 2(i, iii, vi), 3, 4, 5(i, iii), 6(i, ii, iv), 7, 9, 10** on or before **25th February 2024**.

1. Determine convergence or divergence of each of the following series:

- i. $\sum_{n=1}^{\infty} \sqrt{\frac{n}{2(n+1)}}$;
- ii. $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n^2}\right)$;
- iii. $\sum_{n=1}^{\infty} \sqrt{\frac{3n^2+5n+4}{4n^2+1}}$;
- iv. $\sum_{n=1}^{\infty} \left(\frac{1}{n}\right)^{1/n}$
- v. $\sum_{n=1}^{\infty} \left(\frac{2}{(-1)^{n-3}}\right)^n$.

2. Determine the convergence or divergence of the following series:

- i. $\sum_{n=1}^{\infty} \frac{1}{n^2+25}$;
- ii. $\frac{1}{5} + \frac{\sqrt{2}}{7} + \frac{\sqrt{3}}{9} + \frac{\sqrt{4}}{11} + \dots$
- iii. $\sum_{n=1}^{\infty} \frac{1}{n^2} \sin\left(\frac{1}{n^2}\right)$;
- iv. $\sum_{n=2}^{\infty} \frac{n}{n^2+3}$
- v. $\sum_{n=1}^{\infty} (\sqrt{n^4+1} - \sqrt{n^4-1})$

3. Use the ratio test to determine the convergence of each of the following series:

- i. $\sum_{n=1}^{\infty} \frac{n!}{n^n}$
- ii. $\sum_{n=1}^{\infty} \frac{n!}{3 \cdot 5 \cdot 7 \cdots (2n+1)}$
- iii. $\frac{1^2 \cdot 2^2}{1!} + \frac{2^2 \cdot 3^2}{2!} + \frac{3 \cdot 4^2}{3!} + \dots$

4. Use the root test to determine the convergence of the following series:

i. $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$

ii. $\sum_{n=1}^{\infty} \frac{n^{n^2}}{(n+1)^{n^2}}$

iii. $\sum_{n=1}^{\infty} \frac{1.3.5 \dots (2n-1)}{2.4.6 \dots 2n} \frac{x^{2n}}{2n}$

(you should discuss the convergence leading to the values of $x \in \mathbb{R}$.)

5. Use the integral test to determine the convergence or divergence of the following series:

i. $\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$

ii. $\sum_{n=27}^{\infty} \frac{1}{n \log n \log \log n (\log \log \log n)^p}$

iii. $\sum_{n=1}^{\infty} n e^{-n^2}$

6. Determine the absolute convergence and convergence of the following series:

i. $\sum_{n=1}^{\infty} \frac{(-1)^n \cos nx}{n\sqrt{n}}$

ii. $\sum_{n=1}^{\infty} \frac{\cos nx}{n^2}$

iii. $\sum_{n=1}^{\infty} \frac{\sin nx}{n^2}$

iv. $\sum_{n=1}^{\infty} \frac{(-1)^n (n+2)}{2^{n+5}} \cos nx$

7. Suppose that $\sum_{n=1}^{\infty} a_n$ is a series of positive terms a_n which is convergent.

Prove that $\sum_{n=1}^{\infty} a_n^2$ and $\sum_{n=1}^{\infty} \sqrt{a_n a_{n+1}}$ are convergent.

8. Show that the series $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is convergent.

9. Use the comparison test to show that the series $\sum_{n=1}^{\infty} \frac{1}{(n^2+n)^q}$

i. converges if $q > \frac{1}{2}$;

ii. diverges if $q \leq \frac{1}{2}$.

10. Prove the following tests that were in the class:

- i. The comparison test
- ii. Limit form of comparison test
- iii. Ration test
- iv. Limit form of ratio test
- v. Alternative series test.